

Digital Signal Processing Assignment # 1

BSAI – 7th Semester

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Question 1

Given the difference equation:

$$y(n) = -3y(n-1) + 4y(n-2) + x(n) + x(n-1) - 6x(n-2)$$

(a) Transfer Function

Taking Z-transform of both sides:

$$Z\{y(n)\} = -3Z\{y(n-1)\} + 4Z\{y(n-2)\} + Z\{x(n)\} + Z\{x(n-1)\} - 6Z\{x(n-2)\}$$

Using time-shift property:

$$Z\{y(n-k)\} = z^{-k}Y(z), \quad Z\{x(n-k)\} = z^{-k}X(z)$$

$$Y(z) = -3z^{-1}Y(z) + 4z^{-2}Y(z) + X(z) + z^{-1}X(z) - 6z^{-2}X(z)$$

Grouping terms:

$$Y(z)(1 + 3z^{-1} - 4z^{-2}) = X(z)(1 + z^{-1} - 6z^{-2})$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 + z^{-1} - 6z^{-2}}{1 + 3z^{-1} - 4z^{-2}}$$

(b) Poles and Zeros

Numerator:

$$1 + z^{-1} - 6z^{-2} = 0 \Rightarrow z^2 + z - 6 = 0 \Rightarrow (z+3)(z-2) = 0$$

$$\text{Zeros at } z = -3, 2$$

Denominator:

$$1 + 3z^{-1} - 4z^{-2} = 0 \Rightarrow z^2 + 3z - 4 = 0 \Rightarrow (z+4)(z-1) = 0$$

$$\text{Poles at } z = -4, 1$$

(c) Region of Convergence (ROC)

Since system is causal:

$$\text{ROC: } |z| > 4$$

(d) Stability

A discrete-time system is stable if all poles lie inside the unit circle.

$$|z| = 4 > 1 \Rightarrow \text{System is unstable}$$

(e) Type of Filter

Since feedback terms exist:

$$\boxed{\text{IIR Filter}}$$

Question 2

Given:

$$x(n) = \cos(\omega_0 n) u(n)$$

Using Euler identity:

$$\cos(\omega_0 n) = \frac{e^{j\omega_0 n} + e^{-j\omega_0 n}}{2}$$

$$x(n) = \frac{1}{2} (e^{j\omega_0 n} u(n) + e^{-j\omega_0 n} u(n))$$

Z-transform:

$$Z\{e^{\pm j\omega_0 n} u(n)\} = \frac{1}{1 - e^{\pm j\omega_0} z^{-1}}$$

$$\boxed{X(z) = \frac{z(z - \cos \omega_0)}{z^2 - 2z \cos \omega_0 + 1}}$$

ROC

$$\boxed{|z| > 1}$$

Question 3

(a) FFT Stages

$$N = 8 \Rightarrow \log_2(8) = \boxed{3 \text{ stages}}$$

(b) One FFT Application

$$\boxed{\text{Frequency spectrum analysis}}$$

(c) FFT of Given Signal

$$x(n) = \cos(\omega_x n T), \quad \omega_x = 2\pi \left(\frac{f_s}{4} \right), \quad f_s = 1$$

$$\omega_x = \frac{\pi}{2} \Rightarrow x(n) = \cos\left(\frac{\pi n}{2}\right)$$

FFT shows peaks at:

$$\boxed{k = 2, 6}$$

Question 4

General Z-transform:

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

ROC depends on signal nature:

- Right-sided $\Rightarrow |z| > R$
- Left-sided $\Rightarrow |z| < R$
- Two-sided $\Rightarrow R_1 < |z| < R_2$

Question 5

DFT

$$X(k) = \sum_{n=0}^7 x(n)e^{-j2\pi kn/8}$$

IDFT Verification

$$x(n) = \frac{1}{8} \sum_{k=0}^7 X(k)e^{j2\pi kn/8}$$

Original sequence recovered

Question 6

Given:

$$x(n) = a^n u(n), \quad a > 0$$

Z-transform:

$$X(z) = \sum_{n=0}^{\infty} a^n z^{-n} = \frac{1}{1 - az^{-1}}$$

Poles and Zeros

Pole at $z = a$

ROC

$|z| > a$

Type of Signal

- Causal
- Stable if $|a| < 1$

Question 7

(a) Impulse Response

$$h(n) = \mathcal{Z}^{-1}\{H(z)\}$$

(b) Step Response

$$y(n) = h(n) * u(n)$$

(c) Input $x(n) = (0.25)^n u(n)$

$$X(z) = \frac{1}{1 - 0.25z^{-1}}$$

$$Y(z) = H(z)X(z) \Rightarrow y(n) = \mathcal{Z}^{-1}\{Y(z)\}$$

Conclusion

This assignment provided a complete understanding of Z-transform, ROC, stability, FFT, and system analysis. Each problem was solved step-by-step, making the concepts clear for both written exams and viva voce.